



10 TH MATHS GEOMETRY & GRAPH

COMPLETE QUESTION BANK EM 2024-2025

SIMILAR TRIANGLE

Example : 4.10.

Construct a triangle similar to a given triangle **PQR** with its sides equal to $\frac{3}{5}$ of the corresponding sides of the triangle **PQR** (Scale Factor $\frac{3}{5} < 1$).

Example : 4.11.

Construct a triangle similar to a given triangle **PQR** with its sides equal to $\frac{7}{4}$ of the corresponding sides of the triangle **PQR** (Scale Factor $\frac{7}{4} > 1$).

EXERCISE : 4.1

10. Construct a triangle similar to a given triangle **PQR** with its sides equal to $\frac{2}{3}$ of the corresponding sides of the triangle **PQR** (Scale Factor $\frac{2}{3} < 1$).
11. Construct a triangle similar to a given triangle **LMN** with its sides equal to $\frac{4}{5}$ of the corresponding sides of the triangle **LMN** (Scale Factor $\frac{4}{5} < 1$).
12. Construct a triangle similar to a given triangle **ABC** with its sides equal to $\frac{6}{5}$ of the corresponding sides of the triangle **ABC** (Scale Factor $\frac{6}{5} > 1$).
13. Construct a triangle similar to a given triangle **PQR** with its sides equal to $\frac{7}{3}$ of the corresponding sides of the triangle **PQR** (Scale Factor $\frac{7}{3} > 1$).

SINGLE TANGENT & ALTERNATE SEGMENT & TWO TANGENT

Example 4.29

Draw a circle of **radius 3 cm**. Take a point **P** on this circle and draw a **tangent at P**.

Example 4.30

Draw a circle of **radius 4 cm**. At a point **L** on it draw a **tangent** to the circle using the **alternate segment**.

Example 4.31

Draw a circle of **diameter 6 cm** from a point **P**, which is **8 cm** away from its centre. Draw the **two tangents PA** and **PB** to the circle and measure their lengths.

EXERCISE : 4.4

11. Draw a tangent at any point **R** on the circle of **radius 3.4 cm** and centre at **P?**
12. Draw a circle of **radius 4.5 cm**. Take a point on the circle. Draw the tangent at that point using the **Alternate Segment Theorem**.
13. Draw the **two tangents** from a point which is **10 cm** away from the centre of a circle of **radius 5 cm**. Also, measure the lengths of the tangents.
14. Take a point which is **11 cm** away from the centre of a circle of **radius 4 cm** and draw the **two tangents** to the circle from that point.

15. Draw the **two tangents** from a point which is **5 cm** away from the centre of a circle of **diameter 6 cm**. Also, measure the **lengths of the tangents**.
16. Draw a **tangent** to the circle from the point P having **radius 3.6 cm**. and centre O point P is at a **distance 7.2 cm** from the centre.

Construction of a Triangle

Example : 4.17

Construct a ΔPQR in which $PQ = 8 \text{ cm}$, $\angle R = 60^\circ$ and the **Median RG** from R to PQ is **5.8 cm**. Find the length of the **altitude from R to PQ**.

Example : 4.18

Construct a ΔPQR in which such that $QR = 5 \text{ cm}$, $\angle P = 30^\circ$ and the **Altitude** from P to QR is of length **4.2 cm**.

Example : 4.19

Draw a triangle ΔABC of base $BC = 8 \text{ cm}$, $\angle A = 60^\circ$ and the **Bisector** of $\angle A$ meets BC at D such that **BD = 6 cm**.

EXERCISE : 4.2

11. Construct a ΔPQR in which base $PQ = 4.5 \text{ cm}$, $\angle R = 35^\circ$ and the **Median RG** from R to PQ is **6 cm**.
12. Construct a ΔPQR in which $QR = 5 \text{ cm}$, $\angle P = 40^\circ$ and the **Median PG** from P to QR is **4.4 cm**. Find the length of the **Altitude** from P to QR.
13. Construct a ΔPQR in which such that $QR = 6.5 \text{ cm}$, $\angle P = 60^\circ$ and the **Altitude** from P to QR is of length **4.5 cm**.
14. Construct a ΔPQR in which such that $QR = 5.5 \text{ cm}$, $\angle P = 25^\circ$ and the **Altitude** from P to QR is of length **4 cm**.
15. Draw a triangle ΔABC of base $BC = 5.6 \text{ cm}$, $\angle A = 40^\circ$ and the **Bisector** of $\angle A$ meets BC at D such that **BD = 4 cm**.
16. Draw a triangle ΔPQR such that $PQ = 6.8 \text{ cm}$, **vertical angle** is **50°** and the **Bisector** of vertical angle meets the base at D where **PD = 5.8 cm**.

GRAPH OF VARIATION

Example: 3.47: Varshika drew 6 circles with different sizes. Draw a graph for the relationship between the **diameter and circumference** (approximately related) of each circle as shown in the table and use it to find the circumference of a circle when its **diameter is 6 cm**.

Diameter (x) cm	1	2	3	4	5
Circumference (y) cm	3.1	6.2	9.3	12.4	15.5

Example: 3.48 : A bus is travelling at a uniform speed of **50 km / hr**. Draw the **distance time graph** and hence find

- (i) The **constant of variation**.
- (ii) How far will it travel in **90 minutes or 1 ½ hrs?**.
- (iii) The time required to cover a distance of **300 km** from the graph.

Example: 3.49 : A Company initially started with **40 workers** to complete the work by **150 days**. Later it decided to fasten up the work increasing the number of workers as shown below.

Number of workers (x)	40	50	60	75
Number of days (y)	150	120	100	80

- (i) Graph the above data and identify the **type of variation**.
- (ii) From the graph, find the number of days required to complete the work if the company decides to opt for **120 workers?**.
- (iii) If the work has to be completed by **200 days**, how many workers are required?.

Example: 3.48 : Nishanth is the winner in a Marathon race **12 km distance**. He ran at the uniform speed of **12 km / hr** and reached the destination in **1 hour**. He was followed by Aradhana, Jeyanth, Sathya and Swetha with their respective speed of **6 km / hr, 4 km / hr, 3 km / hr and 2 km / hr**. And, they covered the distance in **2 hrs, 3 hrs, 4 hrs and 6 hours** respectively.

Draw the Speed- time graph and use it to find time taken to Kaushik with his speed **2.4 km / hr**.

EXERCISE 3.15

1. A garment shop announces a flat **50 % discount** on every purchase of items for their customers. Draw the graph for the relation between the **Marked Price and the Discount**. Hence find
 - (i) The marked price when a customer gets a **discount of ₹ 3250 (from graph)**.
 - (ii) The discount when the **marked price is ₹ 2500**.
2. Draw the graph of **$xy = 24, x, y > 0$** . Using the graph find,
 - (i) y when $x = 3$ and
 - (ii) x when $y = 6$.
3. Graph the following **linear function $y = \frac{1}{2}x$** . Identify the constant of variation and verify it with the graph. Also find
 - (i) y when $x = 9$ and
 - (ii) x when $y = 7.5$.
4. The following table shows the data about the number of pipes and the time taken to fill the same tank.

No. of. pipes x	2	3	6	9
Time Taken y (in mins)	45	30	15	10

Draw the graph for the above data and hence

- (i) Find the time taken to fill the tank when **five pipes** are used.
- (ii) Find the number of pipes when the **time is 9 minutes**.

5. A School announces that for a certain competitions, the cash price will be distributed for all the participants equally as show below

No. of. Participants (x)	2	4	6	8	10
Amount for each Participants y (in ₹)	180	90	60	45	36

- (i) Find the **constant of variation**.
- (ii) Graph the above data and hence, find how will each participants get if the number of **participants are 12**.
6. A two wheeler parking zone near bus stand charges as below.

Time x (in hours)	4	8	12	24
Amount y (in ₹)	60	120	180	360

Check if the amount charged are in direct variation or in inverse variation to the parking time. Graph the data. Also

- (i) Find the amount to be paid when parking **time is 6 hrs**.
- (ii) Find the parking duration when the amount **paid is ₹ 150**.

QUADRATIC GRAPH

Example: 3.51: Discuss the nature of solutions of the following quadratic equations.

(i) $x^2 + x - 12 = 0$ (ii) $x^2 - 8x + 16 = 0$ (iii) $x^2 + 2x + 5 = 0$

Example: 3.52: Draw the graph of $y = 2x^2$ and hence solve $2x^2 - x - 6 = 0$.

Example: 3.53: Draw the graph of $y = x^2 + 4x + 3$ and hence solve $x^2 + x + 1 = 0$.

Example: 3.54: Draw the graph of $y = x^2 + x - 2$ and hence solve $x^2 + x - 2 = 0$.

Example: 3.55: Draw the graph of $y = x^2 - 4x + 3$ and hence solve $x^2 - 6x + 9 = 0$.

EXERCISE 3.16

1. Graph the following quadratic equations and state the nature of solutions.

(i) $x^2 - 9x + 20 = 0$ (ii) $x^2 - 4x + 4 = 0$ (iii) $x^2 + x + 7 = 0$ (iv) $x^2 - 9 = 0$
 (v) $x^2 - 6x + 9 = 0$ (vi) $(2x - 3)(x + 2) = 0$

2. Draw the graph of $y = x^2 - 4$ and hence solve $x^2 - x - 12 = 0$.

3. Draw the graph of $y = x^2 + x$ and hence solve $x^2 + 1 = 0$.

4. Draw the graph of $y = x^2 + 3x + 2$ and hence solve $x^2 + 2x + 1 = 0$.

5. Draw the graph of $y = x^2 + 3x - 4$ and hence solve $x^2 + 3x - 4 = 0$.

6. Draw the graph of $y = x^2 - 5x - 6$ and hence solve $x^2 - 5x - 14 = 0$.

7. Draw the graph of $y = 2x^2 - 3x - 5$ and hence solve $2x^2 - 4x - 6 = 0$.

8. Draw the graph of $y = (x - 1)(x + 3)$ and hence solve $x^2 - x - 6 = 0$.

PREPARED & TYPED BY

Y. SEENIVASAN . M.Sc, B.Ed (MATHS TEACHER)